* Lens Distortion

* Perspective projection matrix is based on the pinhole model and is unable to capture distortion that a lens introduce

→ Radial Distortion

→ Deviation from ideal pinhole cause by lens curvature.

(1) Barrel Distortion

→ Defect associated with wide angle lenses that causes straight line ben outward

(2) Pincushion Distortion

→ Defect associated with telephoto lense that cause straight line to bend inward.

→ Tangential Distortion:-

→ Second order defect cause by misalignment or defect of the lens

* Modeling lens Distortion

→ Lens distortion can be modeled using a non-linear transformation that maps ideal image cordinates to distorted (observed image cordinates.

$$\begin{bmatrix} x \\ y \end{bmatrix} = L(r) \underbrace{\begin{bmatrix} x_{undist} \\ y_{undist} \end{bmatrix}}_{\text{radial}} + \underbrace{\begin{bmatrix} dx(x_{undist}, y) \\ dy(x_{undist}, y) \end{bmatrix}}_{\text{tangential}}$$

$r \Rightarrow$ Distance from distortion center $(c = 0,0)$

Therefore $r = \sqrt{(x_{undist})^2 + (y_{undist})^2}$

$L(r) \rightarrow$ radial distortion function which is a linear operation defined for +ve r only & approx using Taylor series

$$L(0) = 1, \quad L(r) = 1 + K_1 r^2 + K_2 r^4 + K_3 r^6 \ldots$$

where K_i = additional intrinsic parameter

* The tangential distortion is approximated as

$$\begin{bmatrix} dx(x_{undist}, y_{undist}, r) \\ dy(x_{undist}, y_{undist}, r) \end{bmatrix} = \begin{bmatrix} 2p_1 x_{undist} y_{undist} + p_2(r^2 + 2x_{undist}^2) \\ 2p_1 y_{undist} x_{undist} + p_2(y_{undist})^2 \end{bmatrix}$$

$\rightarrow p_1$ & p_2 are intrinsic parameters

* Image formation with lens distortion

→ lens distortion is applied after the canonical perspective projection
→ complete flow of image formation

① Transform points from WRF to CRF:-

$$C_1M_w = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

② Apply canonical perspective projection

$$\begin{bmatrix} \frac{x_c}{z_c} \\ \frac{y_c}{z_c} \end{bmatrix} = \begin{bmatrix} x_{undist} \\ y_{undist} \end{bmatrix}$$

③ Apply lens distortion - nonlinear mapping

$$L(r) = \begin{bmatrix} x_{undist} \\ y_{undist} \end{bmatrix} + \begin{bmatrix} dx(x_{undist}, y_{undist}, r) \\ dy(x_{undist}, y_{undist}, r) \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

④ transform CRF to IRF

$$A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ku \\ kv \\ k \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$